Assessment Task – Knowledge, Observation, Portfolio (Learner Version)

Assessment Task 2: Final Practical Exam

Section C – Instructions to learners

Task instructions:

This is a hands-on practical assessment. You are allowed to have with you 2 pages of hand-written commands during the assessment. In this Final practical exam, you will be observed by our instructor as you configure all task for the given topology. You need to complete all task to get a satisfactory result in this unit.

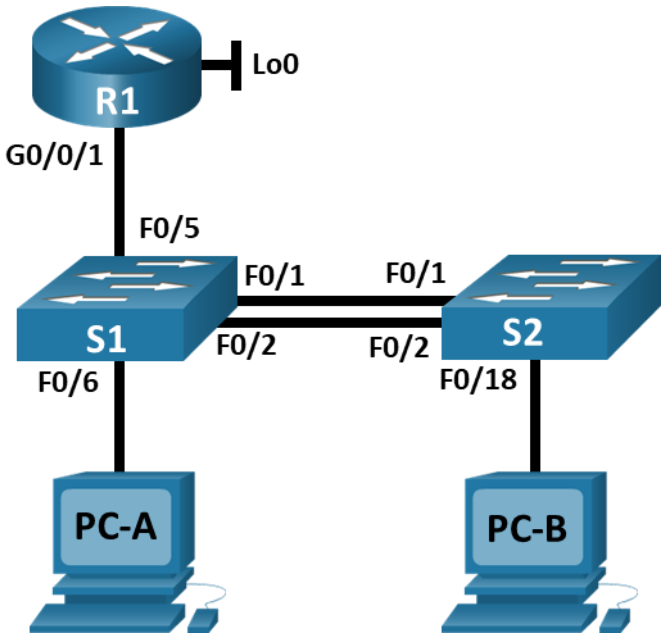
Failing in the first attempt you will be given a second chance to complete this assessment.
It is advisable to practice first configuring the topology using your Packet Tracer.

Students have 2 options on how to complete this assessment. First option is to perform the assessment on campus using the hardware devices. As an incentive for this option, you will be given a grade on CCNA 2 skills assessment exam on Netacad.

The second option is for students attending remotely. You will be performing a video recording of yourself configuring the devices using Packet Tracer. You can use any video recording software to accomplish this.

Every student will submit their Packet Tracer file and Assessment Task 2 document on Brightspace. Remote attending students will also be submitting their recorded video. Your instructor will provide you a feedback on Brightspace within 7 days of submission.

Topology



Assessment Objectives

Part 1: Initialize, Reload and Configure Basic Device Settings

Part 2: Configure Network Infrastructure Settings (VLANs, Trunk, Access)

Part 3: Configure Host Support

Part 4: Test and Verify IPv4 and IPv6 End-to-End Connectivity

Scenario

In this Skills Assessment (SA) you will configure the devices in a small network. You must configure a router, switches, and PCs to support both IPv4 and IPv6 connectivity for supported hosts. Your router and switch must also be managed securely. You will configure inter-VLAN routing, DHCP, access and port-security.

Required Resources

1 Router (Cisco 4221 with Cisco IOS XE Release 16.9.4 universal image or comparable)

2 Switches (Cisco 2960 with Cisco IOS Release 15.2(2) lanbasek9 image or comparable)

2 PCs (Windows with a terminal emulation program, such as Tera Term)

Console cables to configure the Cisco IOS devices via the console ports.

Ethernet cables as shown in the topology.

VLAN Table

VLAN	VLAN Name
2	Bikes
3	Trikes
4	Management
5	Parking
6	Native

Addressing Table

Device / Interface	IP Address / Prefix	Default Gateway
R1 G0/0/1.2	10.19.8.1 /26	N/A
	2001:db8:acad:a::1 /64	N/A
R1 G0/0/1.3	10.19.8.65 /27	N/A
	2001:db8:acad:b::1 /64	N/A
R1 G0/0/1.4	10.19.8.97 /29	N/A
	2001:db8:acad:c::1 /64	N/A
R1 G0/0/1.6	N/A	N/A
R1 Loopback0	209.165.201.1 /27	N/A
	2001:db8:acad:209::1 /64	N/A
S1 VLAN 4	10.19.8.98 /29	10.19.8.97
	2001:db8:acad:c::98 /64	N/A
	fe80::98	N/A
S2 VLAN 4	10.19.8.99 /29	10.19.8.97
	2001:db8:acad:c::99 /64	N/A
	fe80::99	N/A
PC-A NIC	DHCP for IPv4 address	DHCP for IPv4 default gateway
	2001:db8:acad:a::50 /64	fe80::1
PC-B NIC	DHCP for IPv4 address	DHCP for IPv4 default gateway
	2001:db8:acad:b::50 /64	fe80::1

Note: There is no interface on the router supporting VLAN 5.

Instructions

Initialize, Reload and Configure Basic Device Settings

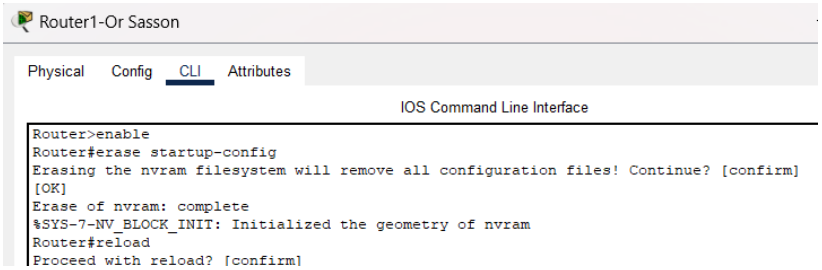
Initialize and reload router and switch.

Erase the startup configurations and VLANs from the router and switch and reload the devices.

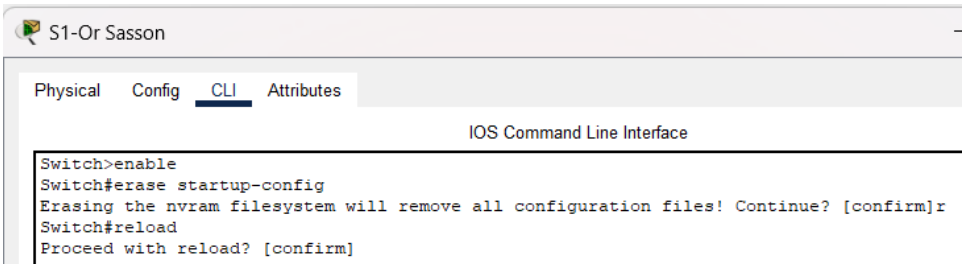
After the switch is reloaded, configure the SDM template to support IPv6 as needed, and reload the switch again.

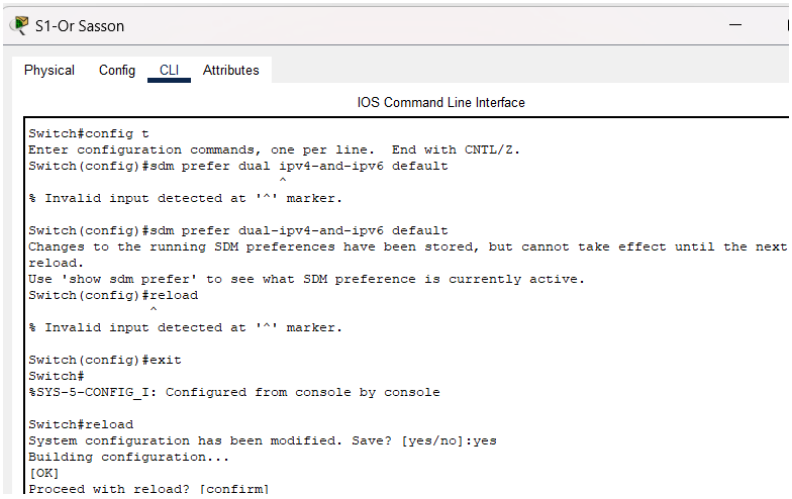
Before proceeding, have your instructor verify device initializations.

R1:

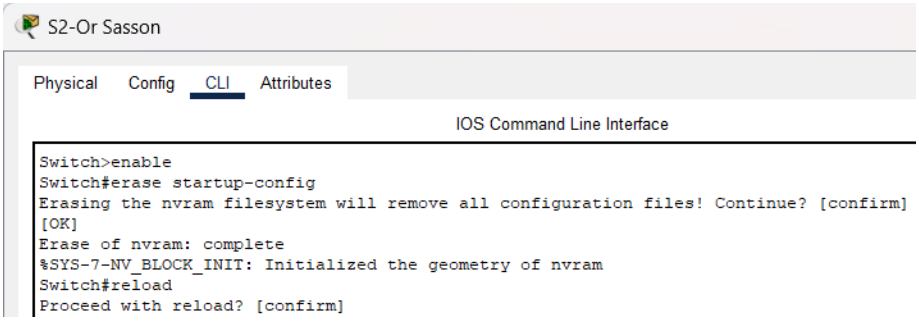


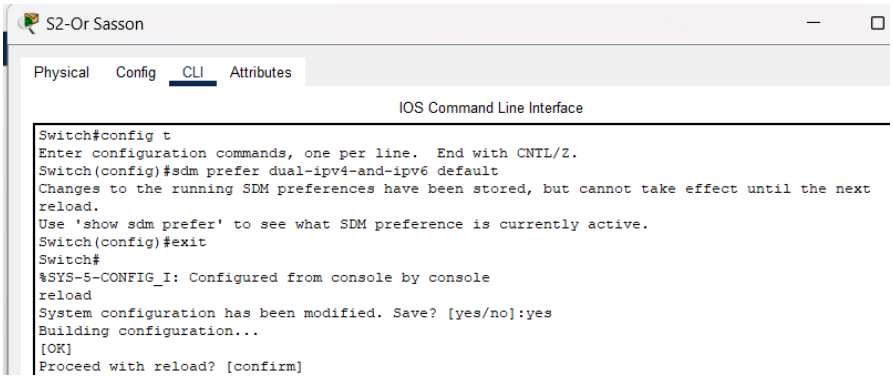
S1





S2



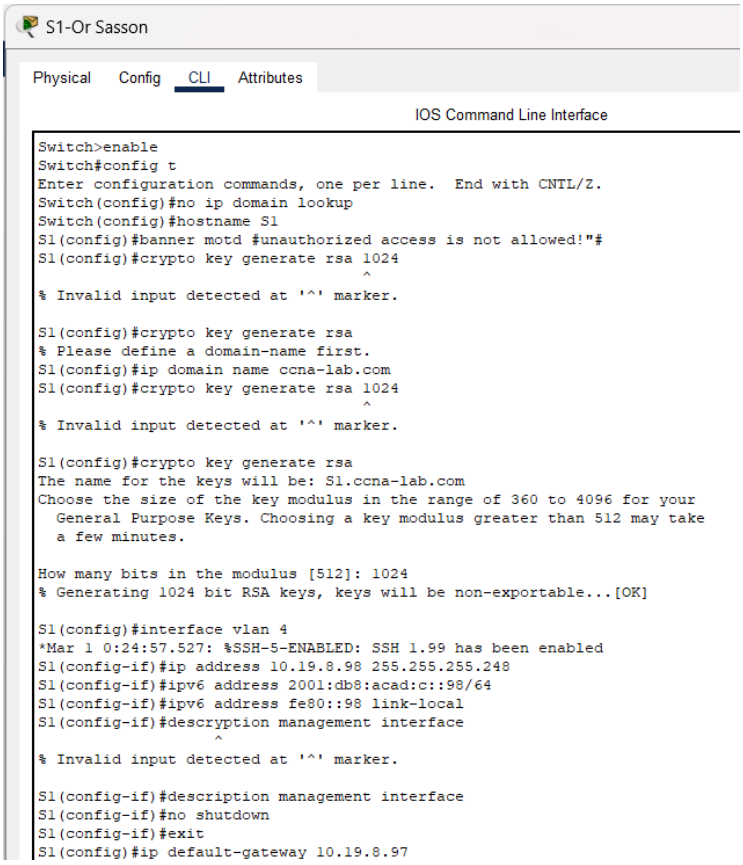


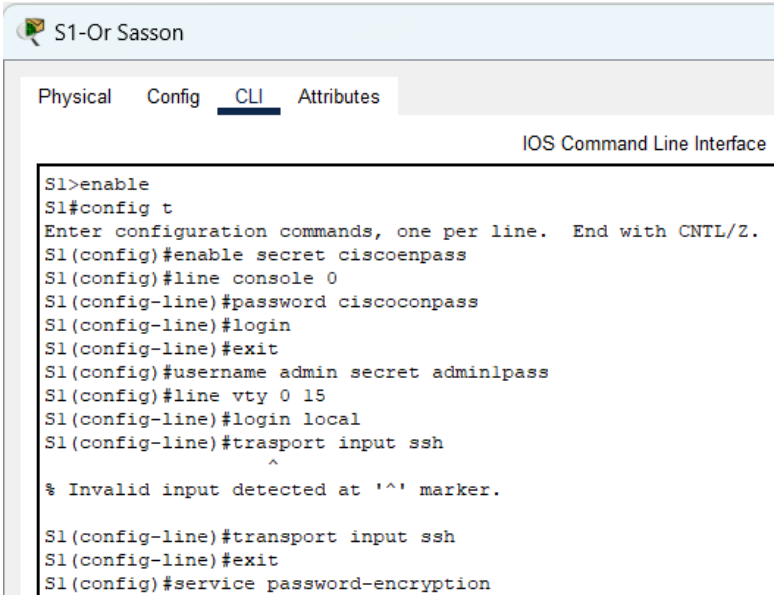
- **Configure S1 and S2.**

Configuration tasks for the switches include the following:

Task	Specification	S1	S2
Disable DNS lookup		Screenshots below	Screenshots below
Switch name	**S1 or S2, as appropriate**		
Domain name	**ccna-lab.com**		
Encrypted privileged EXEC password	**ciscoenpass**		
Console access password	**ciscoconpass**		
Create an administrative user in the local database	Username: **admin** Password: **admin1pass**		
Set login on VTY lines to use local database			
Set VTY lines to accept SSH connections only			
Encrypt the clear text passwords			
Configure an MOTD Banner			
Generate an RSA crypto key	**1024 bits modulus**		
Configure Management Interface (SVI)	Set the Layer 3 IPv4 address Set the Ipv6 Link Local Address as **FE80::98 for S1 and FE80::99 for S2** Set the Layer 3 IPv6 address		
Configure Default Gateway	Configure the default gateway as 10.19.8.97 for IPv4		

S1 configuration:


S1-Or Sasson
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#no ip domain lookup
Switch(config)#hostname S1
S1(config)#banner motd #unauthorized access is not allowed!"#
S1(config)#crypto key generate rsa 1024
% Invalid input detected at '^' marker.
S1(config)#crypto key generate rsa
% Please define a domain-name first.
S1(config)#ip domain name ccna-lab.com
S1(config)#crypto key generate rsa 1024
% Invalid input detected at '^' marker.
S1(config)#crypto key generate rsa
The name for the keys will be: S1.ccna-lab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
S1(config)#interface vlan 4
*Mar 1 0:24:57.527: %SSH-5-ENABLED: SSH 1.99 has been enabled
S1(config-if)#ip address 10.19.8.98 255.255.255.248
S1(config-if)#ipv6 address 2001:db8:acad:c::98/64
S1(config-if)#ipv6 address fe80::98 link-local
S1(config-if)#descryption management interface
% Invalid input detected at '^' marker.
S1(config-if)#description management interface
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#ip default-gateway 10.19.8.97


S1-Or Sasson
Physical
Config
CLI
Attributes
IOS Command Line Interface
S1>enable
S1#config t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#enable secret ciscoenpass
S1(config)#line console 0
S1(config-line)#password ciscoconpass
S1(config-line)#login
S1(config-line)#exit
S1(config)#username admin secret admin1pass
S1(config)#line vty 0 15
S1(config-line)#login local
S1(config-line)#trasport input ssh
% Invalid input detected at '^' marker.
S1(config-line)#transport input ssh
S1(config-line)#exit
S1(config)#service password-encryption


S1-Or Sasson
Physical
Config
CLI
Attributes
IOS Command Line I
S1#show run
Building configuration...
Current configuration : 4022 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname S1
!
!
enable secret 5 1mERr$EJnmB234UvJf9yoQMWYJK/
!
!
!
no ip domain-lookup
ip domain-name ccna-lab.com
!
username admin secret 5 1mERr$ILrAmVhMGbrCFnj8QqS3T.
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel1
 switchport trunk native vlan 6
 switchport mode trunk
!
interface FastEthernet0/1
 switchport trunk native vlan 6
 switchport trunk allowed vlan 2-6
 switchport mode trunk
 channel-group 1 mode active
!
interface FastEthernet0/2
 switchport trunk native vlan 6
 switchport trunk allowed vlan 2-6
 switchport mode trunk
 channel-group 1 mode active

S2 configuration:

S2-Or Sasson

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#no ip domain lookup
Switch(config)#hostname S2
S2(config)#ip domain name ccna-lab.com
S2(config)#banner motd #unauthorized access is not allowed!"#
S2(config)#crypto key generate rsa
The name for the keys will be: S2.ccna-lab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

S2(config)#interface vlan 4
*Mar 1 0:28:31.399: %SSH-5-ENABLED: SSH 1.99 has been enabled
S2(config-if)#ip address 10.19.8.99 255.255.255.248
S2(config-if)#ipv6 address 2001:db8:acad:c::99/64
S2(config-if)#ipv6 address fe80::99 link-local
S2(config-if)#descryption Management Interface
                ^
% Invalid input detected at '^' marker.

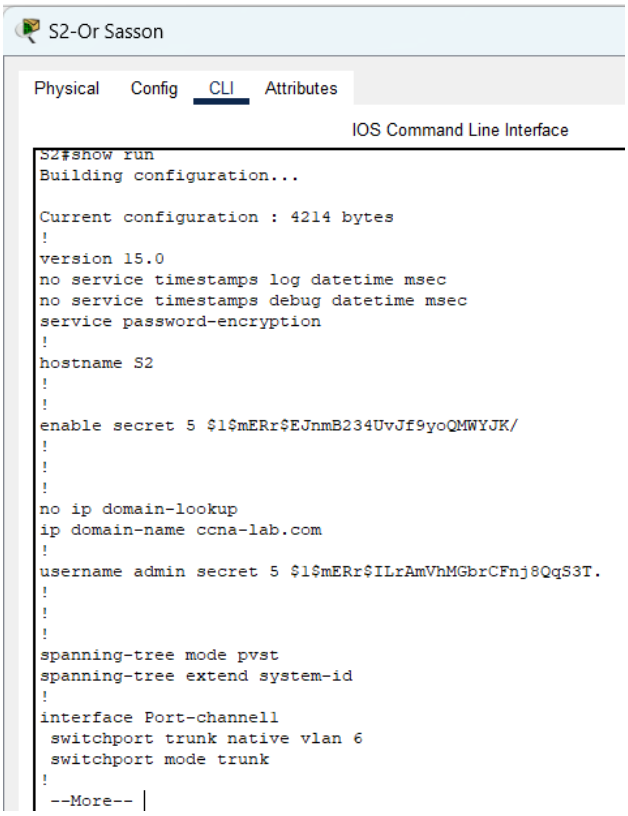
S2(config-if)#description Management Interface
S2(config-if)#no shutdown
S2(config-if)#exit
S2(config)#ip default-gateway 10.19.8.97
```

S2-Or Sasson

Physical Config CLI Attributes

IOS Command Line Interface

```
S2>enable
S2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#enable secret ciscoenpass
S2(config)#line console 0
S2(config-line)#password ciscoconpass
S2(config-line)#login
S2(config-line)#exit
S2(config)#username admin secret adminlpass
S2(config)#line vty 0 15
S2(config-line)#login local
S2(config-line)#transport input ssh
S2(config-line)#exit
S2(config)#service password-encryption
S2(config)#exit
```



Instructor Sign-off Part 1:

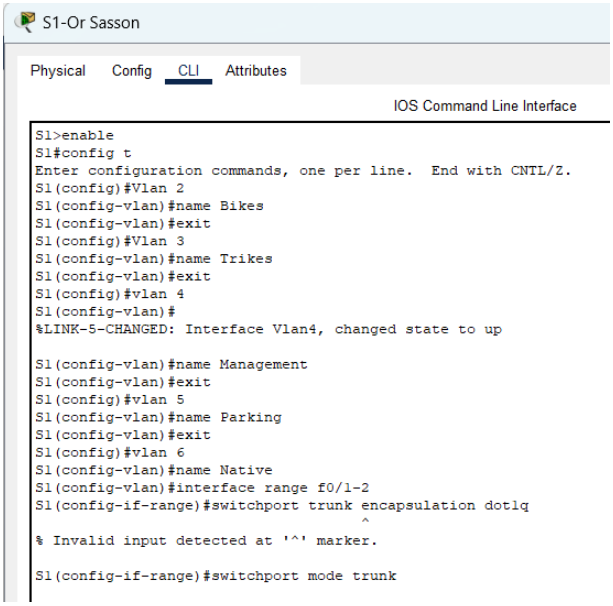
Instructor Sign-off

Configure Network Infrastructure Settings (VLANs, Trunk, Access)

- **Configure S1.**

Configuration tasks for S1 include the following:

Task	Specification	Completed Y/N
Create VLANs	VLAN 2, name Bikes VLAN 3, name Trikes VLAN 4, name Management VLAN 5, name Parking VLAN 6, name Native	Y
Create 802.1Q trunks that use the native VLAN 6	Interfaces F0/1, F0/2, and F0/5	Y
Configure host access port for VLAN 2	Interface F0/6	Y
Configure port-security on access ports	Allow 3 MAC addresses	Y
Secure all unused interfaces	Assign to VLAN 5, Set to access mode, add a description, and shutdown	Y




```
S1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

S1(config-if-range)#switchport trunk native vlan 6
```

```
S1(config-if-range)#switchport trunk allowed vlan 2-6
S1(config-if-range)#exit
S1(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (6), with S2
FastEthernet0/1 (1).

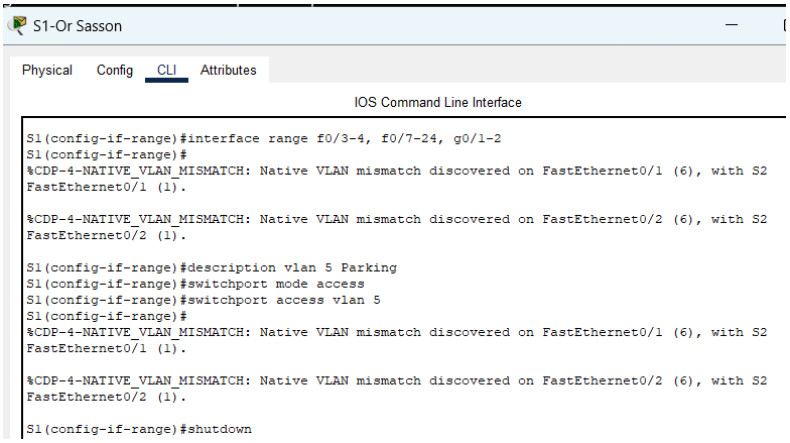
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (6), with S2
FastEthernet0/2 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (6), with S2
FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (6), with S2
FastEthernet0/2 (1).
```

```
S1(config)#interface f0/5
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk native vlan 6
S1(config-if)#
```

```
S1(config)#interface f0/6
S1(config-if)#switchport mode access
S1(config-if)#switchport
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN misma
FastEthernet0/2 (1).
a
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN misma
FastEther
S1(config-if)#switchport access vlan 2
S1(config-if)#switch port-security maximum 3
S1(config-if)#
```



- **I got a message that the VLAN is mismatch. This is normal as I did not configure S2 yet.**

- **Configure S2.**

Configuration tasks for S2 include the following:

Task	Specification	Completed Y/N
Create VLANs	VLAN 2, name Bikes VLAN 3, name Trikes VLAN 4, name Management VLAN 5, name Parking VLAN 6, name Native	Y
Create 802.1Q trunks that use the native VLAN 6	Interfaces F0/1 and F0/2	Y
Configure host access port for VLAN 3	Interface F0/18	Y
Configure port-security on access ports	Allow 3 MAC addresses	Y
Secure all unused interfaces	Assign to VLAN 5, Set to access mode, add a description, and shutdown	Y

```
S2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#vlan 2
S2(config-vlan)#name Bikes
S2(config-vlan)#vlan 3
S2(config-vlan)#name Trikes
S2(config-vlan)#vlan 4
S2(config-vlan)#
%LINK-5-CHANGED: Interface Vlan4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up

S2(config-vlan)#name Management
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with S1
FastEthernet0/1 (6).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with S1
FastEthernet0/2 (6).

S2(config-vlan)#name Management
S2(config-vlan)#vlan 5
S2(config-vlan)#name Parking
S2(config-vlan)#vlan 6
S2(config-vlan)#name Native
```

```
S2(config-vlan)#interface range f0/1-2
S2(config-if-range)#switchport mode trunk
S2(config-if-range)#switchport trunk native vlan 6
```

```
switchport trunk allowed vlan 2-6
S2(config-if-range)#switchport trunk allowed vlan 2-6
S2(config-if-range)#exit
```

```
S2(config-if-range)#exit
S2(config)#int f0/18
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 3
S2(config-if)#switchport port-security maximum 3
S2(config-if)#exit
```

S2-Or Sasson

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

S2(config-if-range)#exit
S2(config)#int f0/18
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 3
S2(config-if)#switchport port-security maximum 3
S2(config-if)#exit
S2(config)#int range f0/3-17, f0/19-24, g0/1-2
S2(config-if-range)#description vlan 5 Parking - unused
S2(config-if-range)#switchport mode access
S2(config-if-range)#switchport access vlan 5
S2(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
```

- **After configuring switch 2 I didn't get the mismatch message again as expected.**

Instructor Sign-off Part 2:

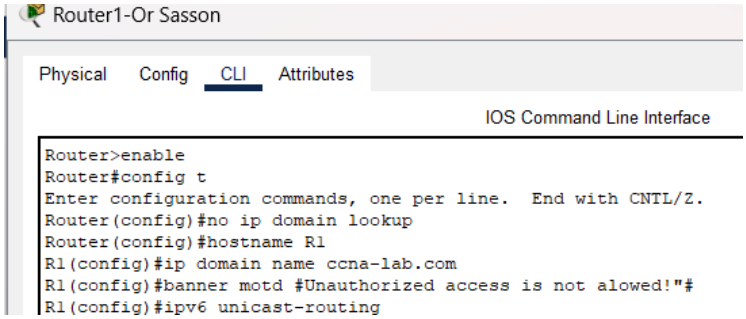
Instructor Sign-off

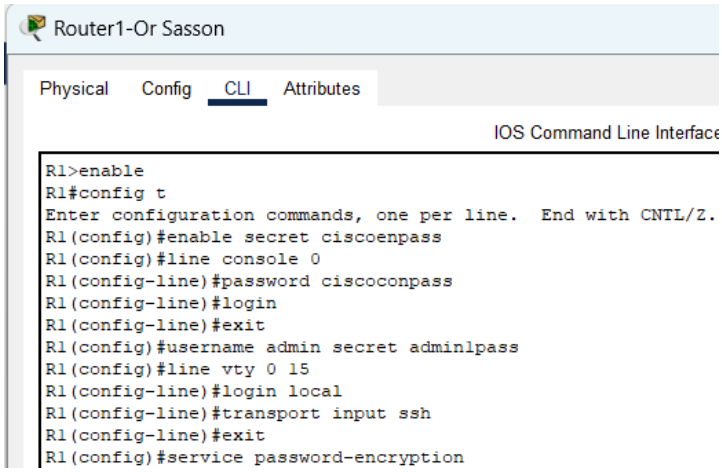
Configure Host Support

- **Configure R1**

Configuration Tasks for R1 include the following:

Task	Specification	Completed Y/N
Configure Default Routing	Create a default route for IPv4 and IPv6 that direct traffic to interface Loopback 0	
Configure IPv4 DHCP for VLAN 2	Create a DHCP pool for VLAN 2, consisting of the last 10 addresses in the subnet only. Assign the domain name ccna-a.net and specify the default gateway address as the router interface address for the subnet involved	
Configure IPv4 DHCP for VLAN 3	Create a DHCP pool for VLAN 3, consisting of the last 10 addresses in the subnet only. Assign the domain name ccna-b.net and specify the default gateway address as the router interface address for the subnet involved	






```
R1(config)#int Gig0/1.2
R1(config-subif)#encapsulation dot1q 2
R1(config-subif)#description vlan 2 Bikes
R1(config-subif)#ip address 10.19.8.1 255.255.255.192
R1(config-subif)#ipv6 address 2001:db8:acad:a::1/64
R1(config-subif)#ip address fe80::1 link-local
                            ^
% Invalid input detected at '^' marker.

R1(config-subif)#ipv6 address fe80::1 link-local
R1(config-subif)#exit
```

```
R1(config)#int g0/1.3
R1(config-subif)#encapsulation dot1q 3
R1(config-subif)#description vlan 3 Trikes
R1(config-subif)#ip address 10.19.8.65 255.255.255.224
R1(config-subif)#ipv6 address 2001:db8:acad:b::1/64
R1(config-subif)#ipv6 address fe80::1 link-local
R1(config-subif)#exit
```

```
R1(config)#int g0/1.4
R1(config-subif)#encapsulation dot1q 4
R1(config-subif)#description vlan 4 Management
R1(config-subif)#ip address 10.19.8.97 255.255.255.248
R1(config-subif)#ipv6 address 2001:db8:acad:c::1/64
R1(config-subif)#ipv6 address fe80::1 link-local
R1(config-subif)#exit
```

```
R1(config)#int g0/1
R1(config-if)#no shutdown
```

```
R1(config)#int loopback 0
R1(config-if)#description Loopback interface
R1(config-if)#ip address 209.165.201.1 255.255.255.224
R1(config-if)#ipv6 address 2001:db8:acad:209::1/64
R1(config-if)#ipv6 address fe80::11 link-local
R1(config-if)#exit
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccna-lab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#
```

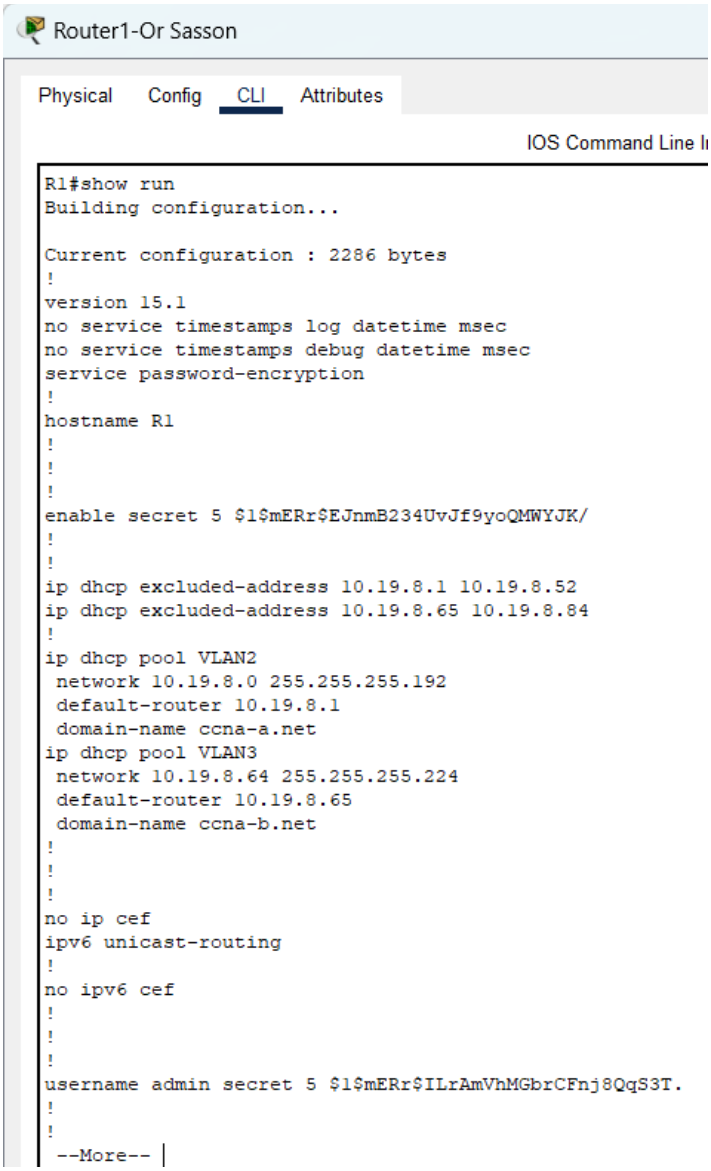
```
R1(config)#ip route 0.0.0.0 0.0.0.0 loopback 0
*Mar 1 2:35:28.233: %SSH-5-ENABLED: SSH 1.99 has been enabled
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#ipv6 route ::/0 loopback 0
```

Because the network of VLAN2 is /26 there are 64 IP addresses available, .63 is the broadcast address so the last 10 addresses are 10.19.8.53 – 10.19.8.62

```
R1(config)#ip dhcp excluded-address 10.19.8.1 10.19.8.52
R1(config)#ip dhcp pool VLAN2
R1(dhcp-config)#network 10.19.8.0 255.255.255.192
R1(dhcp-config)#default-router 10.19.8.1
R1(dhcp-config)#domain-name ccna-a.net
R1(dhcp-config)#exit
```

Because the network of VLAN3 is /27 there are 32 IP addresses available, .95 is the broadcast address so the last 10 addresses are 10.19.8.85 – 10.19.8.94

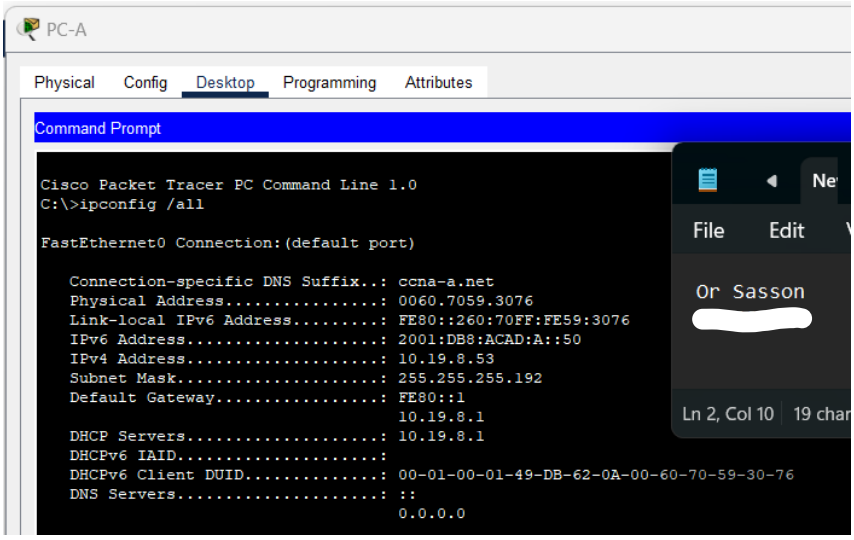
```
R1(config)#ip dhcp excluded-address 10.19.8.65 10.19.8.84
R1(config)#ip dhcp pool VLAN3
R1(dhcp-config)#network 10.19.8.64 255.255.255.224
R1(dhcp-config)#default-router 10.19.8.65
R1(dhcp-config)#domain-name ccna-b.net
R1(dhcp-config)#
```


Router1-Or Sasson
Physical
Config
CLI
Attributes
IOS Command Line I
R1#show run
Building configuration...
Current configuration : 2286 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 1mERr$EJnmB234UvJf9yoQMWYJK/
!
!
ip dhcp excluded-address 10.19.8.1 10.19.8.52
ip dhcp excluded-address 10.19.8.65 10.19.8.84
!
ip dhcp pool VLAN2
network 10.19.8.0 255.255.255.192
default-router 10.19.8.1
domain-name ccna-a.net
ip dhcp pool VLAN3
network 10.19.8.64 255.255.255.224
default-router 10.19.8.65
domain-name ccna-b.net
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username admin secret 5 1mERr$ILrAmVhMGbrCFnj8QqS3T.
!
!
--More--

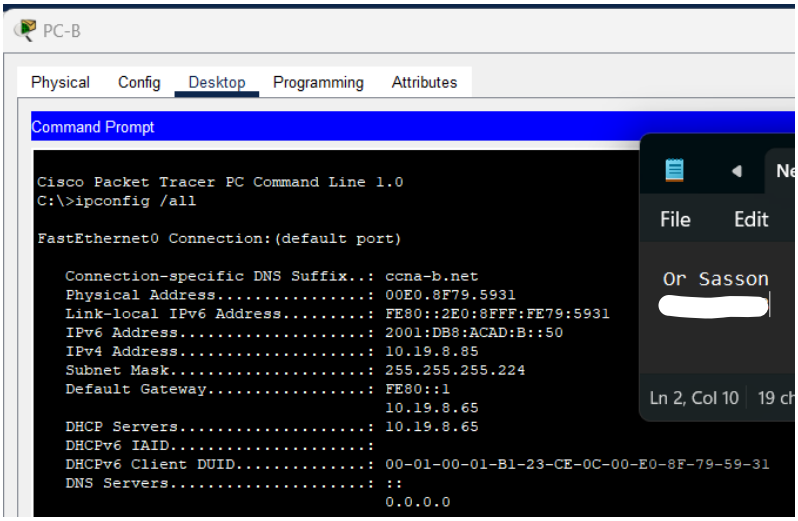
Configure host computers.

Configure the host computers PC-A and PC-B to use DHCP for IPv4 and statically assign the IPv6 GUA and Link Local addresses. After configuring each host computer, record the host network settings with the **ipconfig /all** command.

PC-A Network Configuration	
Description	Attached to VLAN 2
Physical Address	0060.7059.3076
IP Address	10.19.8.53
Subnet Mask	255.255.255.192
Default Gateway	10.19.8.1
IPv6 Default Gateway	FE80::1



PC-B Network Configuration	
Description	Attached to VLAN 3
Physical Address	00E0.8F79.5931
IP Address	10.19.8.85
Subnet Mask	255.255.255.224
Default Gateway	10.19.8.65
IPv6 Default Gateway	Fe80::1



Instructor Sign-off Part 3:

Instructor Sign-off

Test and Verify End-to-End Connectivity

Use the ping command to test IPv4 and IPv6 connectivity between all network devices.

Note: If pings to host computers fail, temporarily disable the computer firewall and retest. (for on-campus)

Use the following table to methodically verify connectivity with each network device. Take corrective action to establish connectivity if a test fails:

From	To	Protocol	IP Address	Ping Results
PC-A	R1, G0/0/1.2	IPv4	10.19.8.1	C:\>PING 10.19.8.1 Pinging 10.19.8.1 with 32 bytes of data: Reply from 10.19.8.1: bytes=32 time<1ms TTL=255 Reply from 10.19.8.1: bytes=32 time<1ms TTL=255 Reply from 10.19.8.1: bytes=32 time<1ms TTL=255 Reply from 10.19.8.1: bytes=32 time<1ms TTL=255 Ping statistics for 10.19.8.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
		IPv6	2001:db8:acad:a::1	C:\>ping 2001:db8:acad:a::1 Pinging 2001:db8:acad:a::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:A::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:A::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:A::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:A::1: bytes=32 time<1ms TTL=255 Ping statistics for 2001:DB8:ACAD:A::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
	R1, G0/0/1.3	IPv4	10.19.8.65	C:\>ping 10.19.8.65 Pinging 10.19.8.65 with 32 bytes of data: Reply from 10.19.8.65: bytes=32 time<1ms TTL=255 Reply from 10.19.8.65: bytes=32 time<1ms TTL=255 Reply from 10.19.8.65: bytes=32 time<1ms TTL=255 Reply from 10.19.8.65: bytes=32 time<1ms TTL=255 Ping statistics for 10.19.8.65: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
		IPv6	2001:db8:acad:b::1	C:\>ping 2001:db8:acad:b::1 Pinging 2001:db8:acad:b::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:B::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:B::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:B::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:B::1: bytes=32 time<1ms TTL=255 Ping statistics for 2001:DB8:ACAD:B::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
	R1, G0/0/1.4	IPv4	10.19.8.97	C:\>ping 10.19.8.97 Pinging 10.19.8.97 with 32 bytes of data: Reply from 10.19.8.97: bytes=32 time=14ms TTL=255 Reply from 10.19.8.97: bytes=32 time<1ms TTL=255 Reply from 10.19.8.97: bytes=32 time<1ms TTL=255 Reply from 10.19.8.97: bytes=32 time<1ms TTL=255 Ping statistics for 10.19.8.97: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 14ms, Average = 3ms

From	To	Protocol	IP Address	Ping Results
		IPv6	2001:db8:acad:c::1	C:\>ping 2001:db8:acad:c::1 Pinging 2001:db8:acad:c::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:C::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:C::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:C::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:C::1: bytes=32 time=1ms TTL=255 Ping statistics for 2001:DB8:ACAD:C::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 1ms, Average = 0ms
	S1, VLAN 4	IPv4	10.19.8.98	C:\>ping 10.19.8.98 Pinging 10.19.8.98 with 32 bytes of data: Reply from 10.19.8.98: bytes=32 time<1ms TTL=254 Reply from 10.19.8.98: bytes=32 time<1ms TTL=254 Reply from 10.19.8.98: bytes=32 time<1ms TTL=254 Reply from 10.19.8.98: bytes=32 time<1ms TTL=254 Ping statistics for 10.19.8.98: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
		IPv6	2001:db8:acad:c::98	Ping isn't successful. After troubleshooting and doing some research I found out that on 2960 switch it isn't possible to config IPv6 route and that's why it cannot ping.
	S2, VLAN 4	IPv4	10.19.8.99	C:\>ping 10.19.8.99 Pinging 10.19.8.99 with 32 bytes of data: Reply from 10.19.8.99: bytes=32 time<1ms TTL=254 Reply from 10.19.8.99: bytes=32 time<1ms TTL=254 Reply from 10.19.8.99: bytes=32 time<1ms TTL=254 Reply from 10.19.8.99: bytes=32 time<1ms TTL=254 Ping statistics for 10.19.8.99: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
		IPv6	2001:db8:acad:c::99	C:\>ping 2001:db8:acad:c::99 Pinging 2001:db8:acad:c::99 with 32 bytes of data: Request timed out. Request timed out. Request timed out. Request timed out. Ping statistics for 2001:DB8:ACAD:C::99: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), Ping isn't successful. After troubleshooting and doing some research I found out that on 2960 switch it isn't possible to config IPv6 route and that's why it cannot ping.

From	To	Protocol	IP Address	Ping Results
	PC-B	IPv4	IP address will vary.	C:\>ping 10.19.8.85 Pinging 10.19.8.85 with 32 bytes of data: Reply from 10.19.8.85: bytes=32 time<1ms TTL=127 Reply from 10.19.8.85: bytes=32 time<1ms TTL=127 Reply from 10.19.8.85: bytes=32 time<1ms TTL=127 Reply from 10.19.8.85: bytes=32 time<1ms TTL=127 Ping statistics for 10.19.8.85: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
		IPv6	2001:db8:acad:b::50	C:\>ping 2001:db8:acad:b::50 Pinging 2001:db8:acad:b::50 with 32 bytes of data: Reply from 2001:DB8:ACAD:B::50: bytes=32 time<1ms TTL=127 Reply from 2001:DB8:ACAD:B::50: bytes=32 time<1ms TTL=127 Reply from 2001:DB8:ACAD:B::50: bytes=32 time<1ms TTL=127 Reply from 2001:DB8:ACAD:B::50: bytes=32 time<1ms TTL=127 Ping statistics for 2001:DB8:ACAD:B::50: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
	R1 Loop0	IPv4	209.165.201.1	C:\>ping 209.165.201.1 Pinging 209.165.201.1 with 32 bytes of data: Reply from 209.165.201.1: bytes=32 time<1ms TTL=255 Reply from 209.165.201.1: bytes=32 time<1ms TTL=255 Reply from 209.165.201.1: bytes=32 time<1ms TTL=255 Reply from 209.165.201.1: bytes=32 time=16ms TTL=255 Ping statistics for 209.165.201.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 16ms, Average = 4ms
		IPv6	2001:db8:acad:209::1	C:\>ping 2001:db8:acad:209::1 Pinging 2001:db8:acad:209::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:209::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:209::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:209::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:209::1: bytes=32 time<1ms TTL=255 Ping statistics for 2001:DB8:ACAD:209::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms

From	To	Protocol	IP Address	Ping Results
PC-B	R1 Loop0	IPv4	209.165.201.1	C:\>ping 209.165.201.1 Pinging 209.165.201.1 with 32 bytes of data: Reply from 209.165.201.1: bytes=32 time<1ms TTL=255 Reply from 209.165.201.1: bytes=32 time<1ms TTL=255 Reply from 209.165.201.1: bytes=32 time<1ms TTL=255 Reply from 209.165.201.1: bytes=32 time<1ms TTL=255 Ping statistics for 209.165.201.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
		IPv6	2001:db8:acad:209::1	C:\>ping 2001:db8:acad:209::1 Pinging 2001:db8:acad:209::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:209::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:209::1: bytes=32 time=1ms TTL=255 Reply from 2001:DB8:ACAD:209::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:209::1: bytes=32 time=1ms TTL=255 Ping statistics for 2001:DB8:ACAD:209::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 1ms, Average = 0ms
	R1, G0/0/1.2	IPv4	10.19.8.1	C:\>ping 10.19.8.1 Pinging 10.19.8.1 with 32 bytes of data: Reply from 10.19.8.1: bytes=32 time<1ms TTL=255 Reply from 10.19.8.1: bytes=32 time<1ms TTL=255 Reply from 10.19.8.1: bytes=32 time<1ms TTL=255 Reply from 10.19.8.1: bytes=32 time<1ms TTL=255 Ping statistics for 10.19.8.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
		IPv6	2001:db8:acad:a::1	C:\>ping 2001:db8:acad:a::1 Pinging 2001:db8:acad:a::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:A::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:A::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:A::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:A::1: bytes=32 time=13ms TTL=255 Ping statistics for 2001:DB8:ACAD:A::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 13ms, Average = 3ms
	R1, G0/0/1.3	IPv4	10.19.8.65	C:\>ping 10.19.8.65 Pinging 10.19.8.65 with 32 bytes of data: Reply from 10.19.8.65: bytes=32 time<1ms TTL=255 Reply from 10.19.8.65: bytes=32 time<1ms TTL=255 Reply from 10.19.8.65: bytes=32 time=1ms TTL=255 Reply from 10.19.8.65: bytes=32 time<1ms TTL=255 Ping statistics for 10.19.8.65: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 1ms, Average = 0ms

From	To	Protocol	IP Address	Ping Results
		IPv6	2001:db8:acad:b::1	C:\>ping 2001:db8:acad:b::1 Pinging 2001:db8:acad:b::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:B::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:B::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:B::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:B::1: bytes=32 time<1ms TTL=255 Ping statistics for 2001:DB8:ACAD:B::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
	R1, G0/0/1.4	IPv4	10.19.8.97	C:\>ping 10.19.8.97 Pinging 10.19.8.97 with 32 bytes of data: Reply from 10.19.8.97: bytes=32 time<1ms TTL=255 Reply from 10.19.8.97: bytes=32 time<1ms TTL=255 Reply from 10.19.8.97: bytes=32 time<1ms TTL=255 Reply from 10.19.8.97: bytes=32 time<1ms TTL=255 Ping statistics for 10.19.8.97: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
		IPv6	2001:db8:acad:c::1	C:\>ping 2001:db8:acad:c::1 Pinging 2001:db8:acad:c::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:C::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:C::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:C::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:C::1: bytes=32 time<1ms TTL=255 Ping statistics for 2001:DB8:ACAD:C::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
	S1, VLAN 4	IPv4	10.19.8.98	C:\>ping 10.19.8.98 Pinging 10.19.8.98 with 32 bytes of data: Reply from 10.19.8.98: bytes=32 time=5ms TTL=254 Reply from 10.19.8.98: bytes=32 time=1ms TTL=254 Reply from 10.19.8.98: bytes=32 time<1ms TTL=254 Reply from 10.19.8.98: bytes=32 time<1ms TTL=254 Ping statistics for 10.19.8.98: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 5ms, Average = 1ms

From	To	Protocol	IP Address	Ping Results
		IPv6	2001:db8:acad:c::98	C:\>ping 2001:db8:acad:c::98 Pinging 2001:db8:acad:c::98 with 32 bytes of data: Request timed out. Request timed out. Request timed out. Request timed out. Ping statistics for 2001:DB8:ACAD:C::98: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), Ping isn't successful. After troubleshooting and doing some research I found out that on 2960 switch it isn't possible to config IPv6 route and that's why it cannot ping.
	S2, VLAN 4	IPv4	10.19.8.99	C:\>ping 10.19.8.99 Pinging 10.19.8.99 with 32 bytes of data: Reply from 10.19.8.99: bytes=32 time<1ms TTL=254 Reply from 10.19.8.99: bytes=32 time<1ms TTL=254 Reply from 10.19.8.99: bytes=32 time<1ms TTL=254 Reply from 10.19.8.99: bytes=32 time=1ms TTL=254 Ping statistics for 10.19.8.99: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 1ms, Average = 0ms
		IPv6	2001:db8:acad:c::99	Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), Ping statistics for 2001:DB8:ACAD:C::99: Request timed out. Request timed out. Request timed out. Request timed out. C:/>ping 2001:db8:acad:c::99 Ping isn't successful. After troubleshooting and doing some research I found out that on 2960 switch it isn't possible to config IPv6 route and that's why it cannot ping.

Instructor Sign-off Part 4:

Instructor Sign-off

Cleanup

NOTE: DO NOT PROCEED WITH CLEANUP UNTIL YOUR INSTRUCTOR HAS GRADED YOUR SKILLS EXAM AND HAS INFORMED YOU THAT YOU MAY BEGIN CLEANUP.

Unless directed otherwise by the instructor, restore host computer network connectivity, and then turn off power to the host computers.

Before turning off power to the router and switch, remove the NVRAM configuration files (if saved) from both devices.

Disconnect and neatly put away all LAN cables that were used in the Final.

Router Interface Summary Table

Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1800	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2801	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
2811	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
4221	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
4300	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.